MMX UTILS

- **Binary Files (for Linux)**
  - mmx.tbc / extract_mmx / gpdci / gxlil / rnmcv / srfdc
- **Requirement**
  - Use RH8.1 (6/12 build or newer)
  - Put extract_mmx / gpdci / gxlil / rnmcv / srfdc in the path
  - Source mmx.tbc in RH command line

Commands

- mmx_dump_dvd
- mmx_add_decap_cell
- mmx_plot_histogram
- mmx_show_lef_pin
- mmx_show_xtor
- mmx_show_decap
- mmx_change_decap
- mmx_show_missing_via
- mmx_clear
- mmx_help

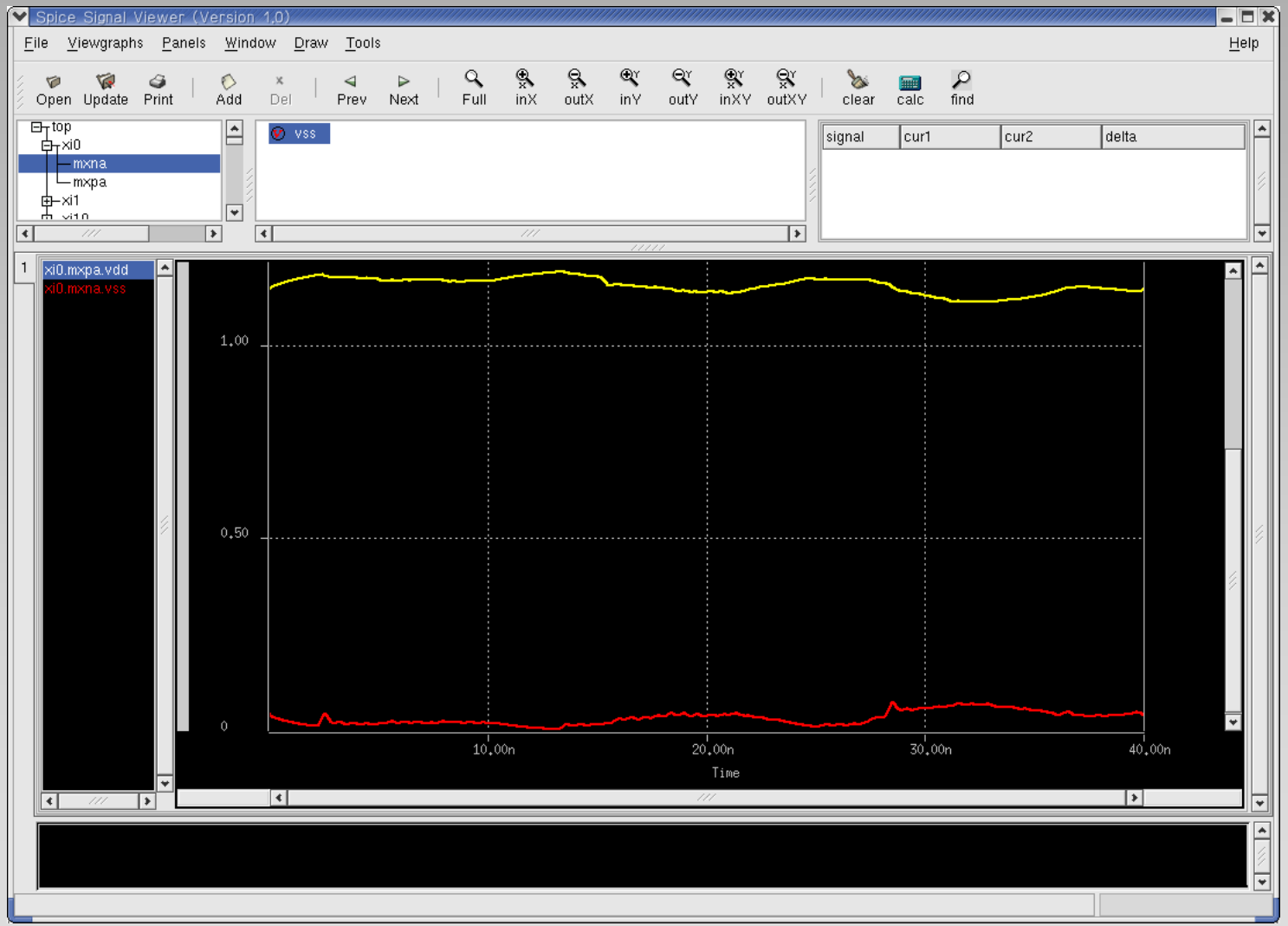
mmx_dump_dvd (1/3)

- **Motivation**
  - **It takes such a long time to see DVD waveforms one by one in a large design**
  - **Users can extract the whole waveforms and see them on a waveform viewer like Sandwork**

mmx_dump_dvd (2/3)

- **Usage**
  - **mmx_dump_dvd –o dvd.ta0**
    - Extract the whole waveforms
  - **mmx_dump_dvd –o dvd.ta0 –index_range 0 99**
    - Extract the worst 100 waveforms (default: 0 ~ 999)
  - **mmx_dump_dvd –o dvd.ta0 –region 100 100 200 200**
    - Extract the waveforms in a certain region
  - **mmx_dump_dvd –o dvd.ta0 –time_base 100n**
    - Specify the start time of capturing window defined in APLMMX configuration
    - The waveforms will begin from 100ns on SV or Sandwork
  - **mmx_dump_dvd –o dvd.ta0 –nocap**
    - Exclude the waveforms of intentional decap xtors

mmx_dump_dvd (3/3)



mmx_add_decap_cell (1/3)

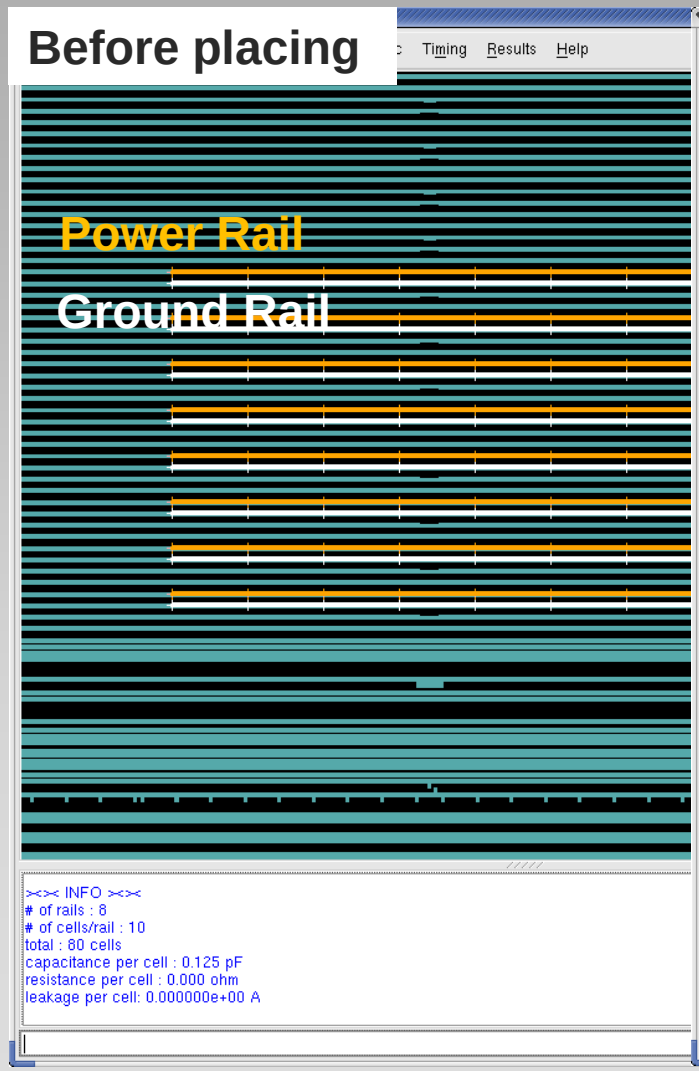
- **Motivation**
  - **‘decap fill’ can not be used in a custom-layout design**
  - **Although ‘Add Decap Cell’ is available, it’s too laborious to place many decap cells manually over a wide region**

mmx_add_decap_cell (2/3)

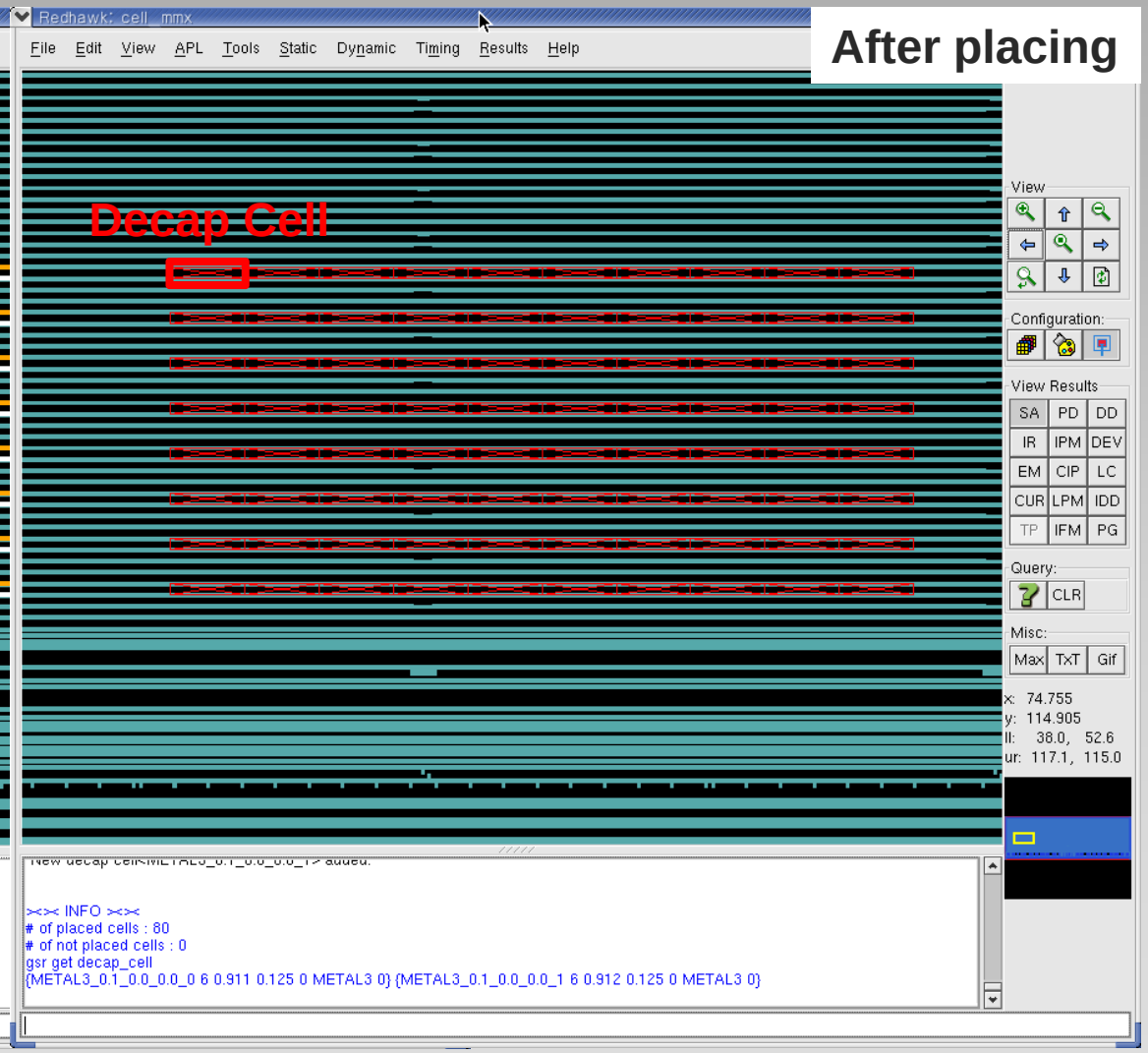
- **Usage**
  - **mmx_add_decap_cell –region 100 100 200 200 –pg vdd vss –metal M1 –cap 10p**
    - Show P/G rails where decap cells can be placed (without '-place')
  - **mmx_add_decap_cell –region 100 100 200 200 –pg vdd vss –metal M1 –cap 10p –place**
    - Make sure the rails are correct and then use '-place' to really place decap cells
  - **mmx_add_decap_cell –region 100 100 200 200 –pg vdd vss –metal M1 –nc 5**
    - Specify the number of decap cells on a rail (default : 10)
  - **mmx_add_decap_cell –region 100 100 200 200 –pg vdd vss –metal M1 –hws 1**
    - Specify a rough value of 0.5x(width+space) (μm units) (default: 0.2)

mmx_add_decap_cell (3/3)

Before placing



After placing



mmx_plot_histogram (1/3)

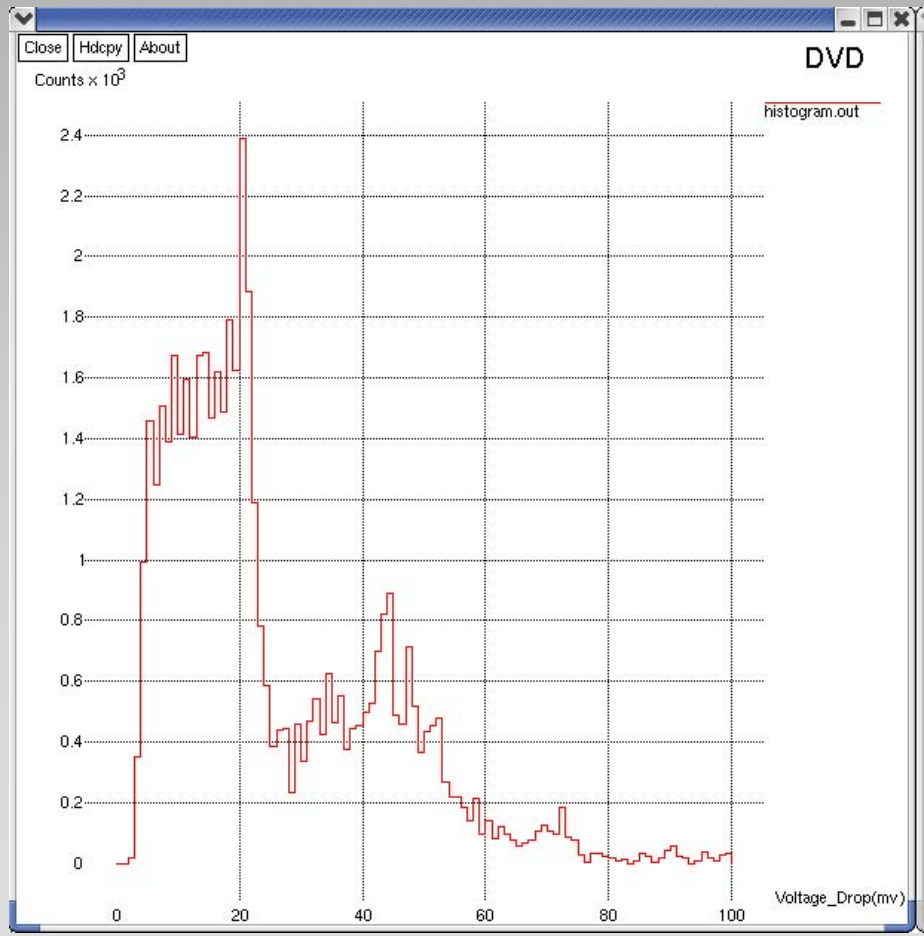
- **Motivation**
  - **Users might want to see a DVD/IR histogram by the number of xtors**

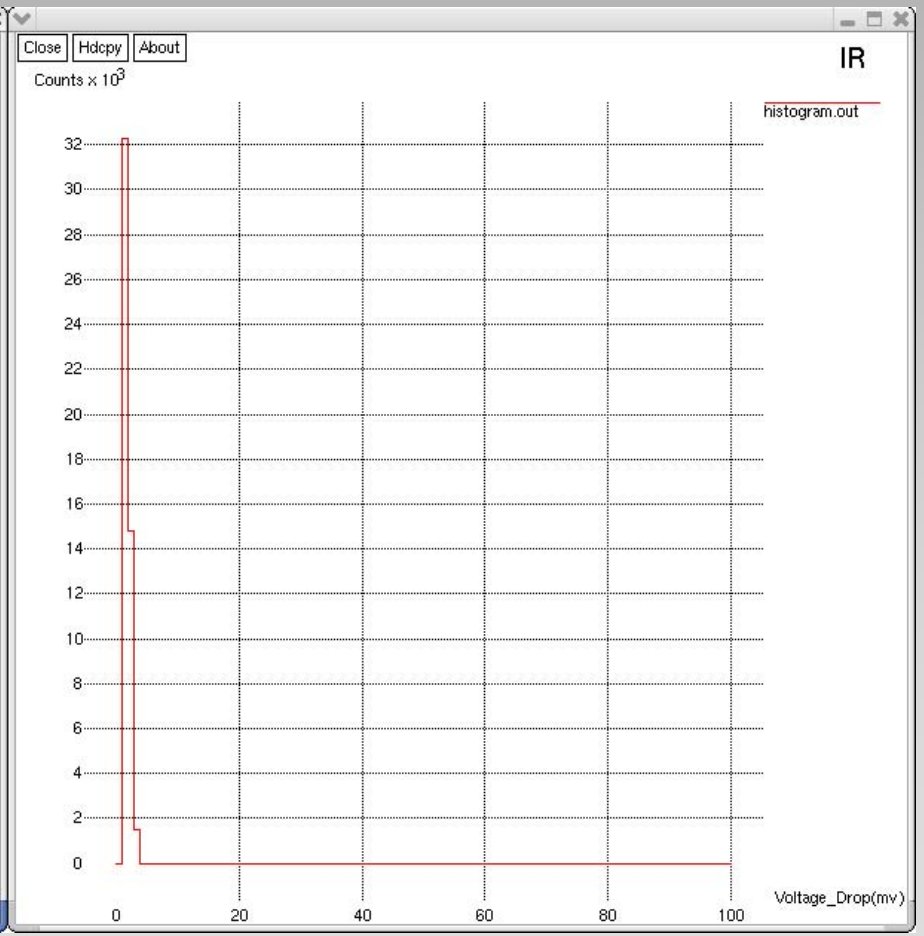
mmx_plot_histogram (2/3)

- **Usage**
  - **mmx_plot_histogram**
    - Plot dynamic voltage drop histogram or static IR histogram
  - **mmx_plot_histogram –bs 10m –pg vdd –log**
    - Set bin size to 10mV, plot histogram of vdd net, and use log scale for Y-axis
  - **mmx_plot_histogram –min 10m –max 300m –o dvd.plot**
    - Specify min/max range of DVD/IR to plot, and specify the output file name
  - **mmx_plot_histogram –type dynamicVectorless/staticIR**
    - Specify the analysis type (dynamicVectorless or staticIR)

mmx_plot_histogram (3/3)

X-axis : Dynamic /IR voltage drop
Y-axis : Transistor numbers





mmx_show_lef_pin (1/3)

- **Motivation**
  - **When BOX_CELLS is used, users might want to see the location of LEF pins of the instances**

mmx_show_lef_pin (2/3)

- **Usage**
  - **mmx_show_lef_pin**
    - Show all LEF pins
  - **mmx_show_lef_pin –cell cellName**
    - Show LEF pins of the specified CELL
  - **mmx_show_lef_pin –inst adsU2**
    - Show LEF pins of the specified instance
  - **mmx_show_lef_pin –region 0 0 100 100**
    - Show LEF pins in the specified region

mmx_show_lef_pin (3/3)

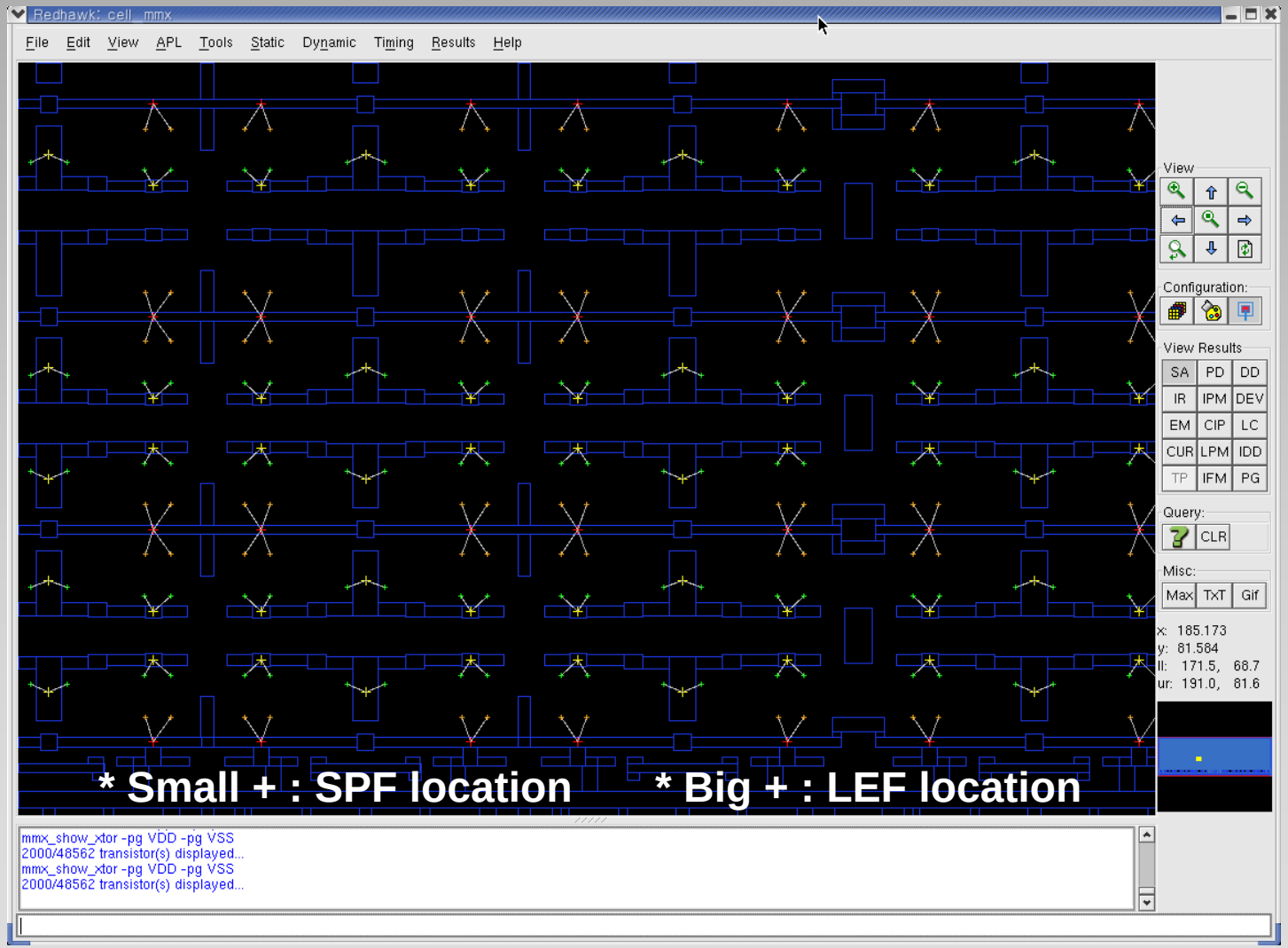
mmx_show_xtor (1/3)

- **Motivation**
  - **Users might want to see if xtors are properly connected to P/G**

mmx_show_xtor (2/3)

- **Usage**
  - **mmx_show_xtor –pg vdd –pg vss –pg vddq –pg vssq**
    - Show xtors connected on vdd, vss, vddq, and vssq
  - **mmx_show_xtor –pg vdd –region 100 100 200 200**
    - Show xtors in a certain region
  - **mmx_show_xtor –pg vdd –nocap**
    - Exclude intentional decap xtors
  - **mmx_show_xtor –pg vdd –n 10000**
    - Specify the max limit of the displayed xtors (default: 1000)

mmx_show_xtor (3/3)



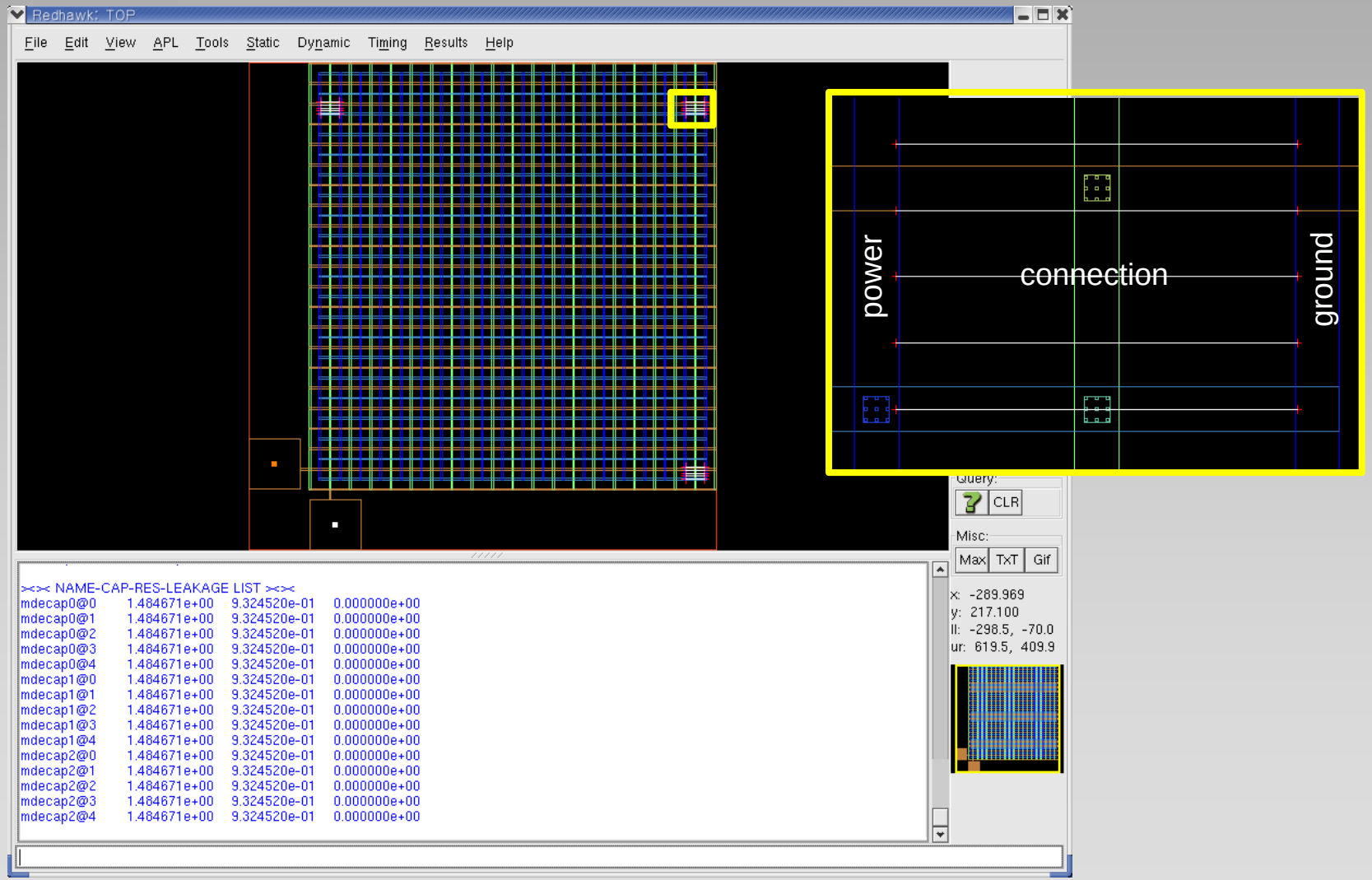
mmx_show_decap (1/3)

- **Motivation**
  - **Users might want to know where intentional decap xtors are residing**

mmx_show_decap (2/3)

- **Usage**
  - **mmx_show_decap –pg vdd vss**
    - Show intentional decap xtors between vdd and vss
    - Red and Blue are Max and Min capacitance respectively
  - **mmx_show_decap –pg vdd vss –region 100 100 200 200**
    - Show intentional decap xtors in a certain region
  - **mmx_show_decap –pg vdd vss –name**
    - Print xtors names
  - **mmx_show_decap –pg vdd vss –conn**
    - Show connection between power and ground of each intentional decap xtor
  - **mmx_show_decap –pg vdd vss –mono**
    - Show intentional decap xtors without coloring (much faster)

mmx_show_decap (3/3)



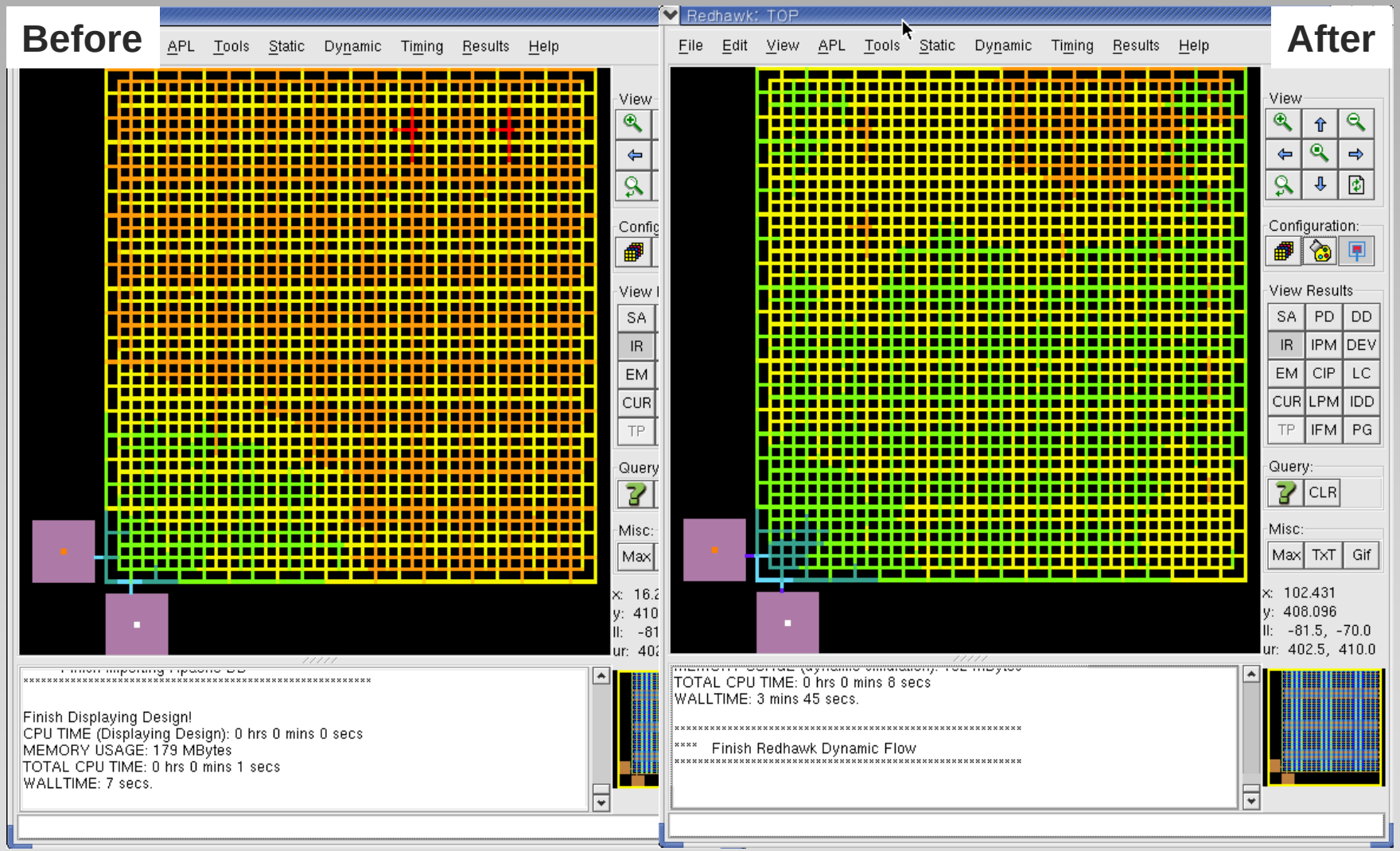
mmx_change_decap (1/3)

- **Motivation**
  - **Changing cdev is another way of decap FAO in MMX**
  - **Users can modify the capacitance of intentional decap xtors and perform dynamic analysis again**

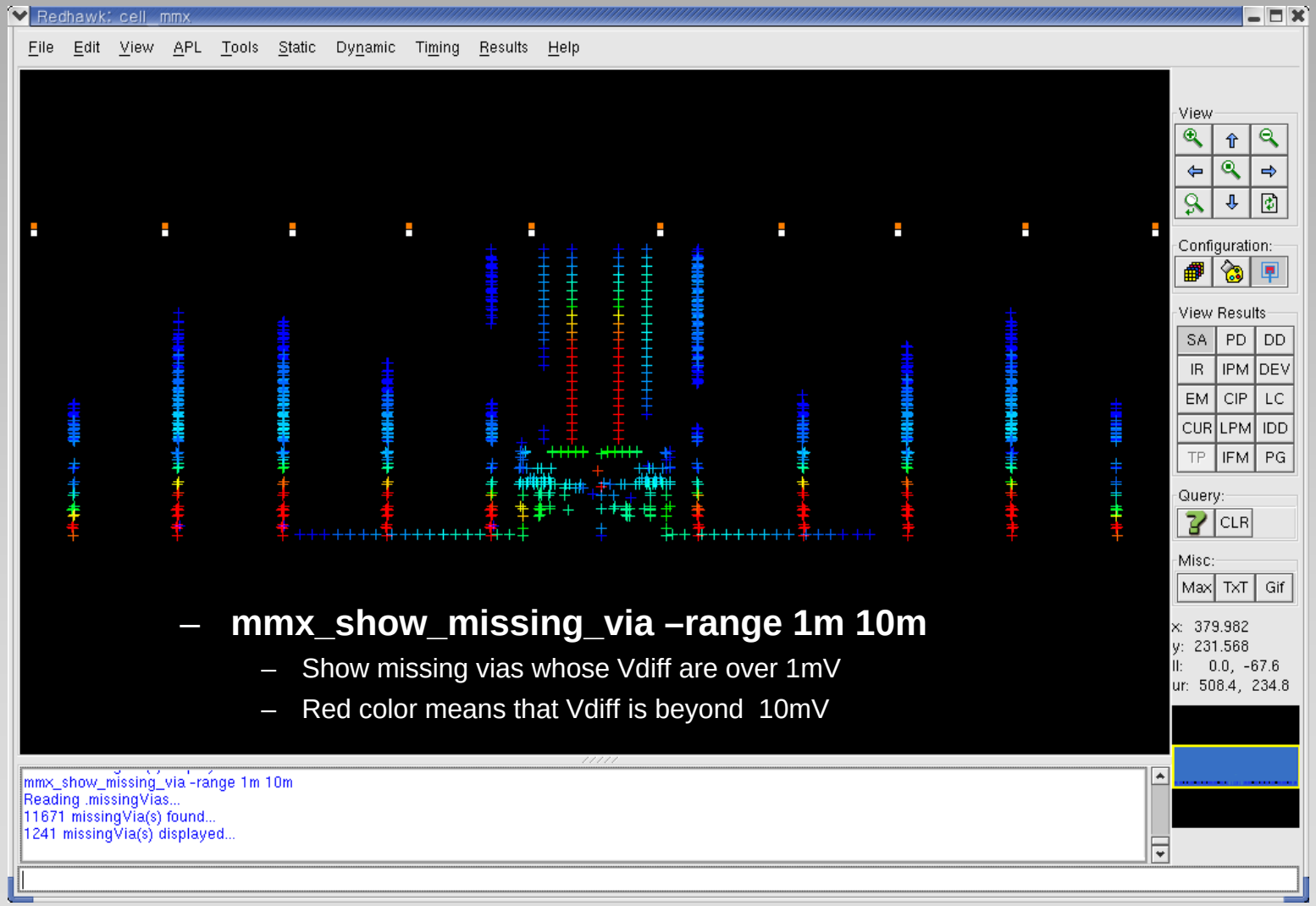
mmx_change_decap (2/3)

- **Usage**
  - **mmx_change_decap –pg vdd vss –pg vddq vssq –region 100 100 200 200 –region 300 300 400 400**
    - Show intentional decap xtors in certain regions
  - **mmx_change_decap –o user.cdev –pg vdd vss –region 0 0 200 200 –cap 3x –change**
    - Triple the capacitance in the region and generate user.cdev
  - **mmx_change_decap –pg vdd vss –region 0 0 200 200 –cap 10p –change**
    - Distribute 10pF to the region and generate custom.cdev
  - **mmx_change_decap –pg vdd vss –region 0 0 200 200 –cap 10p –res 1k –leakage 1u –run**
    - Distribute Cap/Res/Leakage to the region
    - Generate custom.cdev and perform dynamic analysis

mmx_change_decap (3/3)



mmx_show_missing_via



- **mmx_show_missing_via –range 1m 10m**
  - Show missing vias whose Vdiff are over 1mV
  - Red color means that Vdiff is beyond 10mV

mmx_clear / mmx_help

- mmx_clear
  - Clear the screen
- mmx_help
  - Show all mmx_* commands